Firms, Production, and Production Functions

EC306: Intermediate Microeconomics — Day 4

Core reading: Perloff (2020), Ch. 6

Crossing to the supply side

Days 1–3: the consumer chose quantities to maximise utility subject to a budget. Out came *demand*.

Days 4–5: the firm chooses *inputs* to produce output, then chooses output to maximise profit. Out comes *supply*.

The structural parallel — use it

Isoquant ↔ indifference curve. MRTS ↔ MRS. Isocost ↔ budget line. Cost minimisation ↔ utility maximisation. Everything you learned about tangency transfers directly.

Roadmap

1. The firm and the production function
2. Short run vs long run
3. Total, average, and marginal product; diminishing returns
4. Isoquants and the MRTS
5. Returns to scale
6. Functional forms and the markets they describe

The production function: the firm's technology

$$q = f(L, K)$$

maps inputs (labour L, capital K) to the *maximum* output q they can produce.

"Maximum" is doing work

f already assumes the firm is technically efficient — no slack, no waste. The production function is the *frontier*, not a description of what a badly-run firm actually does. Inefficiency is a deviation *below* f.

We treat technology as given today; tomorrow we attach input *prices* and derive costs.

Short run vs long run — a definition about flexibility, not the calendar

Short run (SR)

At least one input is **fixed** (usually $K = \bar{K}$). The firm varies only L.

Long run (LR)

All inputs are variable. The firm can re-scale plant, capital, everything.

Concrete

For a coffee shop, the SR is days–months: you can add baristas (L) but the espresso machines and floor space (K) are fixed. The LR is the horizon over which you can sign a new lease and buy more machines. "Run" length differs wildly across industries — a food truck's LR is weeks; a nuclear plant's is decades.

Total, average, and marginal product (short run)

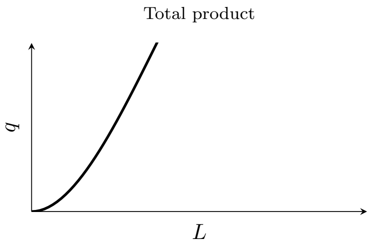
With $K = \bar{K}$ fixed, vary L:

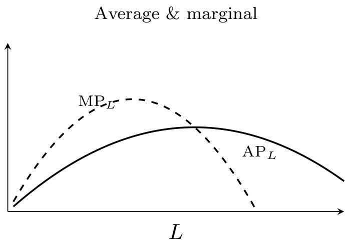
$$\text{AP}_L = \frac{q}{L}, \qquad \text{MP}_L = \frac{\partial q}{\partial L}.$$

Law of diminishing marginal returns

As you add more of a *variable* input to *fixed* inputs, beyond some point MP_L falls. Not a flaw in the worker — it's the fixed capital getting crowded. The 10th cook in a one-oven kitchen adds little.

The TP–AP–MP relationships





Key facts: MP$_L$ peaks first (inflection of TP); MP$_L$ cuts AP$_L$ at the *maximum* of AP$_L$; TP peaks where MP$_L = 0$.
(When marginal > average, average rises — same maths as a test score.)

Your turn (1) — in-lecture

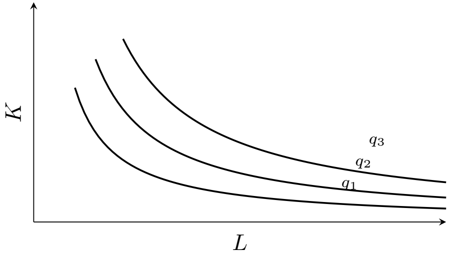
Your turn

$q = 12L^2 - L^3$ (short run, K fixed).

1. Find MP_L and AP_L.
2. At what L does AP_L reach its maximum, and verify $\mathrm{MP}_L = \mathrm{AP}_L$ there.

Isoquants: the firm's indifference curves

An **isoquant** joins all (L, K) combinations yielding the same output q.



Higher isoquants = more output. Their slope is the **MRTS** — the rate at which the firm can swap capital for labour and hold output fixed.

The MRTS

$$\text{MRTS} = -\frac{\mathrm{d}K}{\mathrm{d}L}\bigg|_{q=\bar{q}} = \frac{\text{MP}_L}{\text{MP}_K}.$$

Same derivation as the MRS: totally differentiate $f(L, K) = \bar{q}$ and rearrange.

Reading the number

MRTS $= 3$ means: give up 3 units of capital, add 1 worker, output unchanged. *Diminishing* MRTS (convex isoquant) says: as you substitute toward labour, each extra worker replaces less and less capital — the inputs are imperfect substitutes.

Do not confuse MRTS (a technology object, slope of isoquant) with returns to scale (what happens when you scale *both* inputs). Different questions.

Returns to scale: scale *all* inputs by $t > 1$

Compare $f(tL, tK)$ with $t\,f(L, K)$:

- **Increasing** RTS: $f(tL, tK) > t\,f(L, K)$ — double inputs, more than double output.
- **Constant** RTS: $f(tL, tK) = t\,f(L, K)$.
- **Decreasing** RTS: $f(tL, tK) < t\,f(L, K)$.

Where each shows up

Increasing: software, networks, pipelines (geometry + fixed-cost spreading). **Constant**: often a good benchmark for replicable plants — build an identical second factory. **Decreasing**: management strain, coordination costs, fixed land — why firms don't grow without limit.

Cobb–Douglas: returns to scale from the exponents

For $q = AL^\alpha K^\beta$:

$$f(tL, tK) = A(tL)^\alpha (tK)^\beta = t^{\alpha+\beta} AL^\alpha K^\beta = t^{\alpha+\beta} f(L, K).$$

$$\boxed{\alpha + \beta > 1 : \text{ increasing}; \quad = 1 : \text{ constant}; \quad < 1 : \text{ decreasing.}}$$

Also read off the MRTS

$\text{MP}_L = \alpha AL^{\alpha-1}K^\beta$, $\text{MP}_K = \beta AL^\alpha K^{\beta-1}$, so $\text{MRTS} = \dfrac{\text{MP}_L}{\text{MP}_K} = \dfrac{\alpha}{\beta}\dfrac{K}{L}$ — diminishing in L, as required. One functional form, both properties.

Three more functional forms

Name	Form	Input relationship
Linear / perfect substitutes	$q = aL + bK$	inputs fully interchangeable; straight isoquants
Leontief / fixed proportions	$q = \min\{aL, bK\}$	rigid recipe; L-shaped isoquants
CES	$q = (aL^\rho + bK^\rho)^{1/\rho}$	tunable substitutability; nests the others

Map to real production

Leontief: one driver per truck; one pilot per cockpit — you cannot trade pilots for planes. **Substitutes**: gas vs electric heating for the same heat output. **CES** is the empirical default in macro/growth precisely because ρ lets you *estimate* how substitutable capital and labour really are — a live question for automation.

Your turn (2) — in-lecture

Your turn

$q = 10L^{0.3}K^{0.5}$.

1. State the returns to scale.
2. Find the MRTS.
3. At $(L, K) = (4, 9)$, how many units of K would one extra worker replace?

What you can now do

1. Interpret a production function and distinguish short run from long run by input flexibility.
2. Derive and relate TP, AP, MP, and explain diminishing marginal returns.
3. Work with isoquants and the MRTS, exploiting the parallel to consumer theory.
4. Classify returns to scale — in general and directly from Cobb–Douglas exponents.
5. Match functional forms to real production technologies.

Tomorrow — the payoff

Attach **input prices**. Minimise cost to get the cost function; split it into short- and long-run; then derive the firm's **supply curve**. That completes the market.